EDO STATE COLLEGE OF AGRICULTURE AND NATURAL RESURCES, IGUORIAKHI, NIGERIA

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 104

COURSE TITLE: CROP PROTECTION

CREDIT LOAD: 3 UNITS

AGT 104: CROP PROCTECTION

INTRODUCTION

Crop protection is an important aspect of agriculture that deals with the prevention and control of factors that reduce crop yield and quality. These harmful factors include insect pests, diseases, weeds, and other organisms that attack crops at different stages of growth.

This course is both **theoretical and practical in nature**. The theoretical aspect focuses on understanding the concepts of pests, diseases, weed biology, and principles of control methods. The practical aspect involves hands-on activities such as identification of pests and diseases, field scouting, laboratory isolation of pathogens, and application of control measures.

Overall, crop protection equips learners with the knowledge and skills needed to protect crops effectively, improve productivity, and ensure sustainable agricultural production.

The course aims and objectives are provided below:

Course aim and objectives

On completion of this course, Students should be able to achieve the following objectives:

- Understand the general principles of crop protection.
- Identify plant diseases and understand methods of control.
- Identify insect pests of crops and understand methods of control.
- Identify weeds and understand methods of control.
- Identify nematode pests of crops and understand methods of control
- Identify vertebrate pests of crops and understand methods of control.

GENERAL PRINCIPLES OF CROP PROTECTION

Crop protection is a vital component of agricultural production that focuses on safeguarding crops from damage caused by pests, diseases, and weeds. These harmful agents significantly reduce both the quality and quantity of agricultural produce. Effective crop protection ensures food security, improves farmers' income, and contributes to sustainable agricultural development.

Crop protection involves the application of scientific knowledge, techniques, and management strategies to prevent, control, or eliminate factors that threaten crop growth and productivity. It is an interdisciplinary field that integrates aspects of entomology, plant pathology, weed science, ecology, and agronomy.

1.1 Importance of Crop Protection in Agriculture

The importance of crop protection in agriculture cannot be overemphasized, as crops are constantly exposed to a wide range of biotic and abiotic stresses. Among these, pests (insects, rodents, birds), diseases (caused by fungi, bacteria, viruses), and weeds are the most destructive.

1. **Reduce crop losses**: One of the primary reasons for crop protection is to prevent yield losses. Without adequate protection, a significant portion of crops may be lost either in the field or during storage. In tropical regions, including countries like Nigeria, these losses can be extremely high due to favorable environmental conditions for pest and disease proliferation.

2. **Improve crop quality**: Crop protection also ensures improved quality of produce. Infestation by pests or infection by diseases often results in poor-quality grains, fruits, or vegetables, which may be rejected in markets or fetch lower prices. Therefore, effective crop protection enhances both market value and consumer acceptability.

3. **Increase food security**: Another important aspect is food security. By minimizing losses and increasing productivity, crop protection contributes directly to the availability of food for the growing population.

4. **Supports farmer livelihoods**: It also supports economic stability by improving farm income and reducing the cost of production losses. By reducing crop losses, farmers can maintain their income and livelihoods.

5. **Support sustainable agriculture**: Crop protection plays a role in environmental conservation. When properly managed, it reduces the need for excessive chemical use, thereby minimizing pollution and preserving beneficial organisms such as pollinators and natural enemies of pests., i.e using environmentally friendly and sustainable practices to protect crops while minimizing harm to the environment.

1.2 Various Crop Protection Methods

Crop protection methods refer to the different approaches used to prevent or control pests, diseases, and weeds. These methods are broadly classified into cultural, biological, chemical, mechanical, and quarantine methods.

Cultural Methods

Cultural control involves the use of good farming practices to reduce the incidence of pests and diseases. These practices are preventive in nature and aim to create unfavorable conditions for pest development while promoting healthy crop growth.

Examples include crop rotation, which breaks the life cycle of pests and pathogens; proper spacing, which improves air circulation and reduces disease spread; timely planting, which helps crops escape peak pest periods; and field sanitation, which involves removing crop residues that may harbor pests and pathogens.

Soil management practices such as proper fertilization and irrigation also contribute to crop protection by enhancing plant vigor and resistance.

Biological Methods

Biological control involves the use of living organisms to suppress pest populations. These organisms include predators, parasites (parasitoids), and pathogens that naturally attack pests.

For example, ladybird beetles feed on aphids, parasitic wasps lay eggs inside pest insects, and certain fungi and bacteria infect and kill harmful pests. Biological control is environmentally friendly and sustainable, as it reduces reliance on chemical pesticides.

However, it requires careful management and understanding of ecological relationships to ensure effectiveness.

Chemical Methods

Chemical control involves the use of pesticides such as insecticides, fungicides, and herbicides to control pests, diseases, and weeds. This method is widely used due to its rapid and effective action.

While chemical control can provide immediate results, it must be used judiciously to avoid negative effects such as environmental pollution, pesticide resistance, and harm to non-target organisms including humans.

Proper dosage, timing, and application techniques are essential to maximize effectiveness and minimize risks.

Mechanical Methods

Mechanical control involves the use of physical means to remove or destroy pests and weeds. This includes hand-picking of insects, use of traps, hoeing, slashing, and tillage.

Barriers such as nets and fences may also be used to prevent pest entry. Mechanical methods are simple and environmentally safe but may be labor-intensive and less practical for large-scale farming.

Quarantine Methods

Quarantine involves regulatory measures to prevent the introduction and spread of pests and diseases from one region to another. This is especially important in controlling invasive species.

It includes inspection of imported and exported agricultural materials, restriction of movement of infected plant materials, and enforcement of plant health regulations.

Quarantine helps protect local agriculture from new and potentially devastating pests and diseases.

Basis for Crop Protection

The basis for crop protection lies in understanding the interactions between crops, pests, and the environment. Effective crop protection strategies are built on knowledge of pest biology, life cycles, population dynamics, and environmental factors that influence their development.

Economic considerations also form a key basis. Crop protection decisions are often guided by the concept of economic threshold—the level of pest population at which control measures should be applied to prevent economic loss.

Another important basis is ecological balance. Maintaining a balance between pests and their natural enemies helps in reducing pest outbreaks. Disruption of this balance, often through excessive chemical use, can lead to pest resurgence and resistance.

Preventive measures are generally preferred over curative actions, as they are more cost-effective and sustainable in the long term.

1.3 Concept of Integrated Pest Management (IPM)

Integrated Pest Management (IPM) is a holistic and sustainable approach to crop protection that combines different control methods to manage pest populations in an economically and environmentally sound manner.

Integrated Pest Management can be defined as:

A systematic approach to pest control that combines multiple management strategies and practices to reduce pest populations to economically acceptable levels while minimizing risks to human health, the environment, and non-target organisms.

IPM is based on the principle that complete eradication of pests is neither necessary nor desirable. Instead, the goal is to keep pest populations below levels that cause economic damage.

It involves regular monitoring of pest populations, identification of pests and natural enemies, and application of appropriate control measures only when necessary.

IPM integrates cultural, biological, mechanical, and chemical methods in a compatible manner. For example, a farmer may use crop rotation (cultural), introduce natural predators (biological), apply traps (mechanical), and use pesticides only as a last resort (chemical).

The approach emphasizes the use of selective pesticides that target specific pests while minimizing harm to beneficial organisms.

2.0 IDENTIFICATION OF PLANT DISEASES AND METHODS OF CONTROL

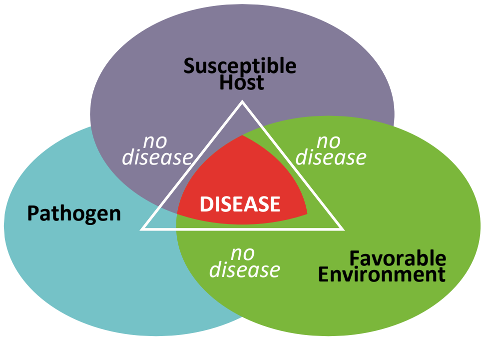
Plant diseases are a major constraint to agricultural productivity, especially in tropical regions such as Nigeria, where climatic conditions favor the rapid growth and spread of pathogens. Understanding plant diseases, their causes, symptoms, modes of transmission, and control measures is essential for effective crop protection and sustainable agricultural production.

2.1 Definition of Disease in Relation to Crops

In crop production, a disease can be defined as any abnormal condition in a plant that disrupts its normal physiological processes, leading to reduced growth, poor yield, or death. This condition is usually caused by pathogenic organisms or unfavorable environmental factors.

A plant disease occurs when there is a continuous interaction between a susceptible host plant, a virulent pathogen, and a conducive environment. This relationship is often referred to as the **disease triangle**. When all three components are present, disease development is inevitable.

The Disease Triangle



The three criteria are present (pathogen, susceptible host and favourable environment)

2.2 Common Diseases of Annual and Tree Crops in Nigeria

In Nigeria, both annual crops (such as maize, rice, cassava, and vegetables) and tree crops (such as cocoa, oil palm, citrus, and mango) are affected by various diseases.

In maize, common diseases include maize streak disease, rust, and leaf blight. Cassava is affected by cassava mosaic disease and cassava bacterial blight. Rice suffers from blast disease and bacterial leaf blight.

Tree crops such as cocoa are affected by black pod disease and swollen shoot disease. Oil palm is susceptible to bud rot and fusarium wilt, while citrus plants commonly suffer from citrus canker and tristeza disease.

These diseases significantly reduce crop yield and quality, leading to economic losses for farmers.

Types of plant diseases

Tens of thousands of diseases affect cultivated and wild plants. On average, each kind of crop plant can be affected by a hundred or more plant diseases. Some pathogens affect only one variety of a plant. Other pathogens affect several dozen or even hundreds of species of plants.

Plant diseases are sometimes grouped according to;

- the symptoms they cause (root rots, wilts, leaf spots, blights, rusts, smuts),
- the plant organ they affect (root diseases, stem diseases, foliage diseases), or
- the types of plants affected (field crop diseases, vegetable diseases, turf diseases, etc.).
- One useful criterion for grouping diseases is the type of pathogen that causes the disease. The advantage of such a grouping is that it indicates the cause of the disease, which immediately suggests the probable development and spread of the disease and also possible control measures.

On this basis, plant diseases in this text are classified as follows:

I. Infectious, or biotic, plant diseases:

a. Diseases caused by fungi

b. Diseases caused by Oomycetes

c. Diseases caused by prokaryotes (bacteria and mollicutes)

d. Diseases caused by parasitic higher plants and green algae

e. Diseases caused by viruses and viroids

f. Diseases caused by nematodes

II. Noninfectious, or abiotic, plant diseases:

a. Diseases caused by too low or too high a temperature

b. Diseases caused by lack or excess of soil moisture

c. Diseases caused by lack or excess of light

d. Diseases caused by lack of oxygen

e. Diseases caused by air pollution

f. Diseases caused by nutrient deficiencies

g. Diseases caused by mineral toxicities

h. Diseases caused by soil acidity or alkalinity (pH)

i. Diseases caused by the toxicity of pesticides

j. Diseases caused by improper cultural practice

2.3 Diseases Caused by Fungi, Bacteria, and Viruses (and Their Vectors)

Plant diseases are broadly classified based on the type of pathogen responsible.

1. FUNGAL DISEASES

Definition

Fungal diseases are plant diseases caused by **fungi**.

Fungi are small, generally microscopic, eukaryotic usually filamentous, branched, spore bearing organisms that lack chlorophyll. Some fungi are referred to as **obligate parasites or biotrophs**. These can grow and multiply only by remaining, during their entire life, in association with their living host plants. Others are known as **non-obligate parasites**. They require a host plant for part of their life cycle but can complete their cycles on dead organic matter, or they can grow and multiply on dead organic matter as well as on living plants. Fungi that are non-obligate parasites can be **facultative saprophytes** or **facultative parasites** depending on whether they are primary parasites or primary saprophytes.

Most fungi have a filamentous vegetative body called a **mycelium**. The mycelium branches out in all directions. The individual branches of the mycelium are called **hyphae** and are generally uniform in thickness, usually about 2 to 10 micrometers in diameter, but in some fungi may be more than 100 um thick. A fungus reproduces by means of spores. Spores are reproductive bodies consisting of one or a few cells.

Only those with cell wall containing chitin are classified as true fungi.
Most plant diseases are caused by fungi and most of them tend to become a problem, especially during the wet weather season.

Some of fungal diseases can cause up to 100% loss in crops

Fungi are the most common cause of plant diseases. They reproduce through spores and thrive in warm, humid environments. They usually enter plants through natural openings (stomata, wounds) and thrive under **warm and humid conditions**.

Characteristics of Fungal Diseases

- Spread by spores (air, water, tools)
- Often produce visible growth (mycelium, mold)
- Common in humid environments
- Can attack leaves, stems, roots, and fruits

Examples of Fungal Diseases

- **Rice blast** (*Magnaporthe oryzae*)
- **Leaf spot of groundnut** (*Cercospora spp.*)
- **Anthracnose of cocoa and mango** (*Colletotrichum gloeosporioides*)
- **Cassava leaf spot** (*Cercospora henningsii*)
- **Fusarium wilt of tomato and banana** (*Fusarium oxysporum*)
- **Powdery mildew of crops** (*Erysiphe spp.*)

- **Black pod disease of cocoa** (*Phytophthora spp.*)

Fungal pathogens often spread through wind, water, soil, or infected plant materials.

***Erysiphales*—Powdery mildews**

Powdery mildews are among the most common, conspicuous, and easily recognized plant diseases. They affect all kinds of flowering plants but not the gymnosperms. The visible symptom of powdery mildew diseases is the mass of asexual spores produced mainly on leaves but also on other aboveground tissues such as stems and fruits. The spores are white to gray and can be easily dislodged by wind or plant movement producing a shower of dry powder, similar to talc. Entire leaves and other organs can be completely covered (Figs. 13.36 and 13.37).

Powdery mildew is most common on the upper side of leaves, but it also affects the underside of leaves, young shoots and stems, buds, flowers, and young fruit (Table 13.8). Fungi causing powdery mildews are obligate parasites and so they cannot be cultured on artificial nutrient media no matter what supplements are added.

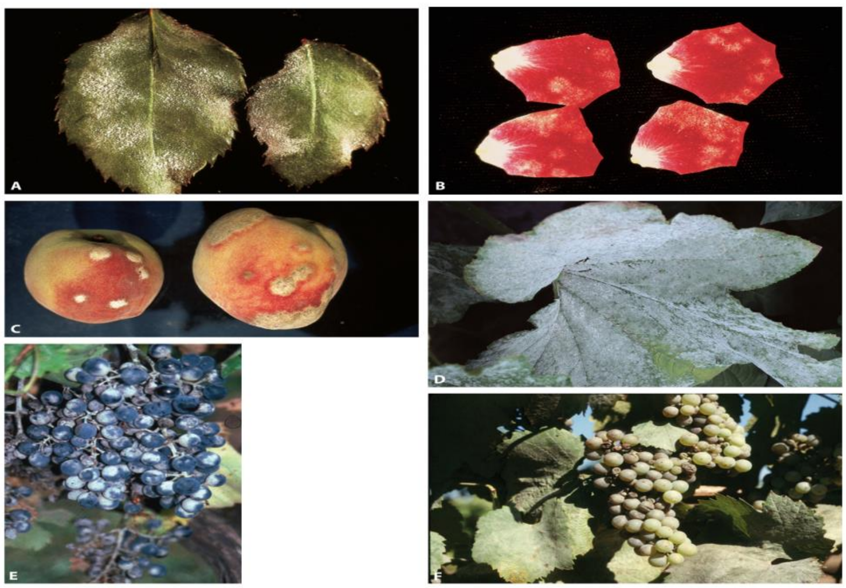


FIG. 13.36 Powdery mildew symptoms on rose leaves (A) and petals (B), peach fruit (C), and squash leaf (D). Powdery mildew symptoms on dark (E) and white (F) grape bunches. *(Photo credits: Plant Pathol. Department, University of Florida (D); J. Travis, Pennsylvania State University (E); and E. Hellman, Texas A&M University (F).)*

Fusarium oxysporum

F. oxysporum is an asexual species complex, often abbreviated the FOSC, which causes disease in more than 120 plant species but also includes pathogens of mammals, insects, as well as saprobes. No sexual cycle has ever been observed. Like *Verticillium,* they cause wilt diseases of many dicot species by invading and blocking the xylem tissues.

The pathogen is a soil inhabitant. Between crops it survives in infected plant debris in the soil as mycelium and in all its spore forms but, most commonly, especially in the cooler temperate regions, as chlamydospores.

It spreads over short distances by means of water and contaminated farm equipment and over long distances primarily by infected transplants or the soil carried with them. Usually, once an area becomes infested with *Fusarium*, it remains so indefinitely.

When healthy plants grow in contaminated soil, the germ tube of spores or the mycelium penetrates root tips directly or enters the roots through wounds or at the point of formation of lateral roots. The pathogen invades the root tissue from mycelium in the soil.

Vessel clogging (Fig. 13.52B, C, and G) by mycelium, spores, gels, gums, and tyloses and crushing of the vessels by proliferating adjacent parenchyma cells, is responsible for the breakdown of the water economy of the infected plant. When the leaves transpire more water than the roots and stem can transport to them, the stomata close and the leaves wilt and finally die, followed by death of the rest of the plant. The fungus then invades all tissues of the plant extensively, reaches the surface of the dead plant, and there sporulates profusely.

FIG. 13.52 *Fusarium* wilt of tomato caused by *Fusarium oxysporum* f. sp. *lycopersici*: wilted tomato plants in the field (A) and severe *brown* discoloration of vascular tissues along the stem of the infected plant (B). Clogged and discolored vascular tissues in cross section of watermelon stem infected with *F. oxysporum* f. sp. *niveum* (C) and infected watermelon plants wilted and dead in the field (D). (E–G) *Fusarium* wilt (Panama disease) of banana caused by *F. oxysporum* f. sp. *cubense*. (E) Lower leaves of infected banana plants wilt, turn brown, and die. (F) Entire infected banana plant killed by the infection. (G) Discolored vascular tissues of infected banana rhizomes. *(Photo credits: R.J. McGovern (A and B); B.D. Bruton (C and D); B. Niere, provided by D. Coyne, IITA, Ibadan, Nigeria (E); and A. Silagyi, University of Florida (F and G).)*

2. BACTERIAL DISEASES

Definition

Bacterial diseases are plant diseases caused by **bacteria**.

Bacteria are **unicellular, microscopic, prokaryotic organisms** that lack a true nucleus and other membrane-bound organelles. In agriculture, plant-pathogenic bacteria invade plant tissues through wounds or natural openings and multiply rapidly, causing diseases such as wilts, blights, soft rots, and leaf spots.

Bacteria thrive in **warm, moist conditions** and often cause rapid tissue decay.

Characteristics of Bacterial Diseases

- Spread through water/rain splash, insects, and infected planting materials/tools
- Cause water-soaked lesions, wilting, and rotting

- Produce foul smell in infected tissues
- Multiply very quickly inside plant tissues

Examples of Bacterial Diseases

- **Bacterial wilt of tomato and pepper** (*Ralstonia solanacearum*)
- **Bacterial leaf blight of rice** (*Xanthomonas oryzae*)
- **Citrus canker** (*Xanthomonas axonopodis*)
- **Soft rot of vegetables (e.g. cabbage, carrots)** (*Erwinia carotovora*)
- **Bacterial leaf spot of pepper** (*Xanthomonas campestris*)

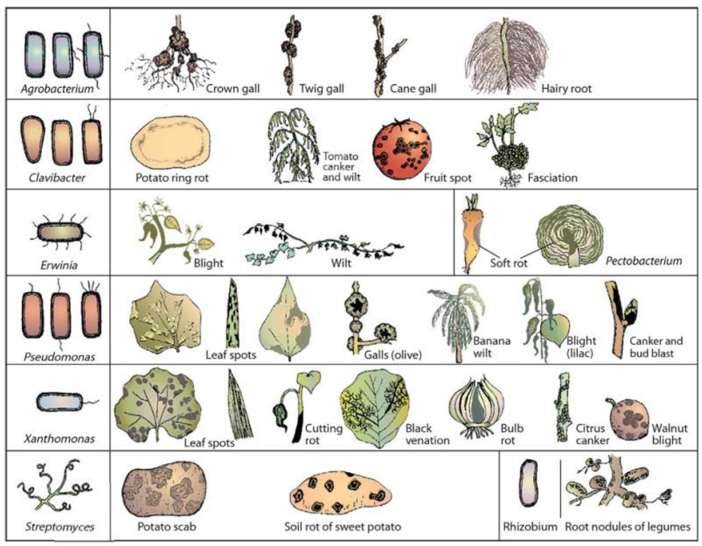
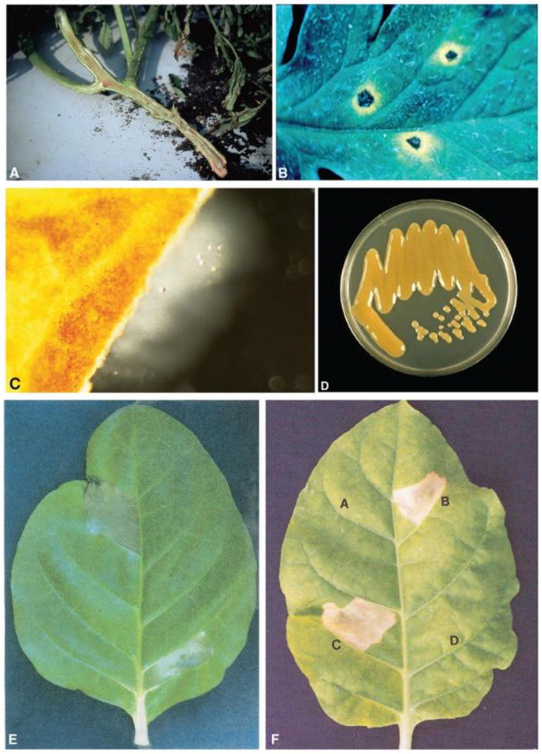


FIG. 16.4 The most important genera of plant pathogenic bacteria and the kinds of symptoms they cause.



IG. 16.5 Some macroscopic features used to approximately determine the bacterial nature of the cause of a plant disease. (A) Brown discoloration of ascular tissues of the wilting plant. (B) The halo surrounding lesions on a leaf of the plant. (C) Cloud-like exudate of bacteria oozing out from infected lant section placed in water. (D) Appearance characteristics of a culture and colonies of bacteria isolated from an infected plant. Hypersensitive reaction ests in which injection of pathogenic bacteria into a leaf of an appropriate nonhost plant induces at first water soaking (E) and then collapse and necrosis of lant tissues (E and F), whereas injection of water at the opposite sides of the leaf at (E) and at (A) of leaf (F) or of a nonpathogenic bacterium (D) at leaf F) induces no such reaction. *(Photo credits: University of Florida, Panels A and C; R.J. McGovern, University of Florida, Panel B; T.R. Gottwald, USDA, t. Pierce, FL, Panel D; and A. Chatterjee, University of Missouri, Panels E and F.)*

3. VIRAL DISEASES

Definition

Viral diseases are plant diseases caused by **viruses**.

A **virus** is a nucleoprotein that multiplies only in living cells and has the ability to cause disease. It is too small to be seen individually with a light microscope. It is made up of genetic material (**DNA or RNA**) enclosed in a protective protein coat called a **capsid**, and in some cases, an outer lipid envelope.

In crop protection, plant viruses are **obligate parasites**, meaning they cannot survive or multiply outside a living plant cell. They invade plant tissues and disrupt normal cellular activities, leading to diseases such as mosaic, stunting, leaf curling, and chlorosis.

Viruses are commonly transmitted by **insect vectors (e.g. aphids, whiteflies), infected planting materials, and mechanical means**.

Characteristics of Viral Diseases

- Cannot be seen with the naked eye or light microscope

- Require living cells to reproduce
- Spread through insect vectors and infected seeds or cuttings
- Cause systemic infection (affects whole plant)

Examples of Viral Diseases

- **Cassava mosaic disease** (Cassava Mosaic Virus – CMV)
- **Maize streak disease** (Maize streak virus)
- **Tobacco mosaic disease** (Tobacco mosaic virus – TMV)
- **Banana bunchy top disease** (Banana bunchy top virus)
- **Pepper veinal mottle disease**

Diseases caused by Virus

Diseases caused by tobamoviruses:

Tobacco mosaic Named after the tobacco mosaic virus, the genus Tobamo virus contains more than a dozen rod-shaped viruses measuring 18-300nm.

There are two closely related viruses of economic importance in the genus: these are tobacco mosaic virus (TMV),which infects tobacco and many other, mostly solanaceous, hosts; and tomato mosaic virus (ToMV), which infects tomato.

Tobamoviruses cause serious losses in their hosts by damaging the leaves, flowers, and fruits and by causing stunting of the plant. The losses are greatest when the plants are infected young. Infections at later stages of growth cause smaller losses. Tobamoviruses are easily transmitted mechanically, and in nature they are spread by incidental contact and wounding, often by cultivation equipment. They do not seem to be transmitted by any vectors. Symptoms consist of various degrees of mottling, chlorosis, curling, distortion, and dwarfing of leaves (Fig. 17.29), flowers, and entire plants.

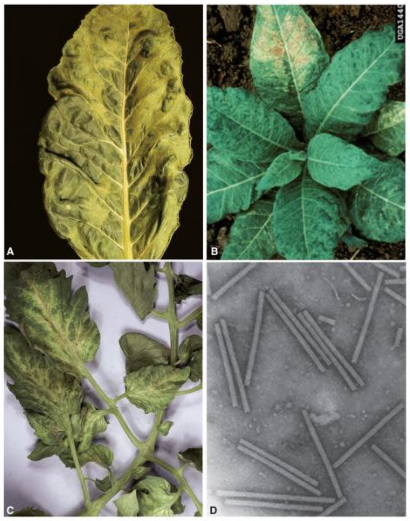


FIG. 17.29 Symptoms of tobacco mosaic infection on (A) tobacco leaf, (B) tobacco plant, and (C) tomato leaves. (D) Particles of the tobamovirus tobacco mosaic virus. *(Photo credits: A and C: Plant Pathology Department, University of Florida; B: E.J. Reynolds Co.)*

Sugarcane mosaic

Sugarcane mosaic occurs worldwide, wherever sugarcane is grown. Its many strains also infect corn (maize), sorghum, and other Gramineae. The disease can be very severe. Symptoms appear as pale patches or blotches on the leaves, not of uniform width and not confined between the veins. Stems may show mottling or marbling, the affected areas later becoming necrotic. The stems become small and deformed; the shoots remain stunted and produce a few twisted or distorted leaves. Cane and sugar yields are reduced severely.

Sugarcane mosaic virus (750 11nm) is transmitted primarily vegetatively in sugarcane during the propagation of the crop. In sugarcane and in all other grain crops, however, the sugarcane mosaic virus is also transmitted by several aphid species in a non-persistent manner. The virus over seasons in infected sugarcane or inappropriate perennial hosts of the specific strains. Control is possible only through the use of resistant or tolerant varieties.

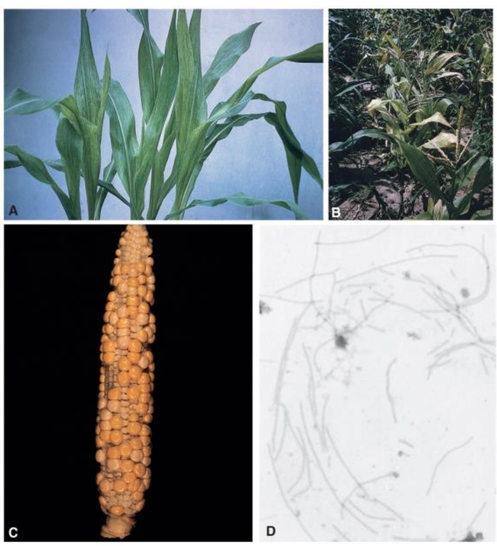


FIG. 17.37 Maize dwarf mosaic on corn caused by the maize strain of the sugarcane mosaic virus (SCMV). (A) Mosaic on young leaves of corn plant. (B) Mosaic, yellowing-reddening and stunting of the corn plant. (C) Poorly filled ear of corn from SCMV-infected plant. (D) Electron micrograph of the virus.

SUMMARY COMPARISON

Pathogen Type	Definition	Mode of Spread	Examples of Diseases
Fungi	Spore-producing organisms that feed on plants	Air, water, soil	Rice blast, anthracnose, powdery mildew
Bacteria	Single-celled microorganisms causing tissue decay	Water, insects, tools	Bacterial wilt, citrus canker
Viruses	Sub-microscopic agents that reproduce in plant cells	Insects, planting materials	Cassava mosaic, maize streak

IMPORTANT DIFFERENCE

Organism	Nature	Cell Type	Nutrition	Example Disease
Fungi	Eukaryotic microorganisms	Eukaryote	Absorptive (heterotrophic)	Anthracnose
Bacteria	Single-celled organisms	Prokaryote	Absorptive	Bacterial wilt
Nematodes	Microscopic worms (animals)	Eukaryote (animal)	Feeding with stylet	Root-knot disease

Description of Pathogens and Their Plant Hosts

Each pathogen has specific host plants that it infects. For instance, *Phytophthora spp.* primarily infect cocoa pods, causing black pod disease, while *Xanthomonas spp.* infect cassava and citrus plants.

Viruses such as cassava mosaic virus specifically target cassava plants, leading to severe yield reduction. Similarly, rice blast fungus affects rice plants at different growth stages.

Understanding host-pathogen relationships helps in developing targeted control strategies and selecting resistant crop varieties.

2.4 Effects, Symptoms, and Spread of Plant Diseases

Effects of Plant Diseases

Plant diseases reduce crop productivity by interfering with essential processes such as photosynthesis, respiration, and nutrient uptake. Severe infections may lead to total crop failure.

Diseases also affect the quality of produce, making them unmarketable or unsafe for consumption. In some cases, toxins produced by pathogens can pose health risks to humans and animals.

Symptoms of Plant Diseases

Symptoms vary depending on the pathogen and crop but commonly include:

- Leaf discoloration (yellowing or chlorosis)
- Wilting of plant parts

- Necrosis (death of tissues)
- Stunted growth
- Leaf spots, blights, and rots
- Deformation of plant organs
- Mosaic patterns on leaves (common in viral infections)

These symptoms help in diagnosing the type of disease affecting a crop.

Spread of Plant Diseases

Plant diseases spread through various means, including:

- Wind (carrying fungal spores)
- Water (rain splash, irrigation)
- Soil (harboring pathogens)
- Infected seeds and planting materials
- Insects and other vectors
- Human activities (tools, machinery, movement of crops)

NEMATODE PESTS OF CROPS

3.1 Definition of Nematodes

Nematodes are microscopic, worm-like, unsegmented invertebrate animals belonging to the phylum Nematoda. They are commonly referred to as roundworms. While some nematodes are beneficial, many are plant parasites that live in the soil and attack crop roots.

Plant-parasitic nematodes possess specialized mouthparts called stylets, which they use to pierce plant cells and extract nutrients. They are usually invisible to the naked eye but cause significant damage to crops.

3.2 Common Nematode Pests Affecting Crops

Several nematodes attack crops in Nigeria and other tropical regions. The most important include:

- Root-knot nematodes (*Meloidogyne spp.*)
- Cyst nematodes (*Heterodera spp.*)
- Lesion nematodes (*Pratylenchus spp.*)
- Stem and bulb nematodes (*Ditylenchus spp.*)
- Reniform nematodes (*Rotylenchulus spp.*)

Among these, root-knot nematodes are the most widespread and destructive, affecting crops such as tomato, okra, maize, and cassava.

3.3 Modes of Infection, Symptoms, and Damage Caused by Nematodes

Mode of Infection

Nematodes infect plants primarily through the roots. They penetrate root tissues using their stylets and establish feeding sites within the plant.

KINDS OF PARASITISM IN PLANT NEMATODE.

Plant-parasitic nematode punctures plant tissue when in contact with the plant.

The puncturing process through intact surfaces is made possible by repeated thrusts (pressing) of tissue by the stylet. Several hundred species of nematodes are known to affect plants. Every crop has its full compliment of nematodes which attack them. Below and above ground parts roots, bush, stem, floor are usually parasitized by the nematodes. Most of the plant of the plants parasitic nematode feed on plant roots. Depending on the part of the plants attacked plant-parasitic nematodes may be classified as follows:

a) **ROOT PARASITES**: This can further be classified as:

i) **ENDOPARASITES**: These are plant parasite nematodes which enters plant roots completely with their entire body. They therefore stays, develops, mature and lay eggs within the root. Endoparasites may be

1) Sessile (sedentary endoparasites) which are those that become stationery as soon as they penetrate the root and pick up their feeding sites. E.g. *Meloidogyne* spp, *Globodera, Heterodera* spp, *Nacobbus* spp. etc. They are most virulent of all plant parasitic nematode. They are responsible for galling of roots. Only 2nd stages are infective.

2) Migratory Endoparasites which normally advances (move) from one cell to the other. E.g. *Pratylenchus Penetrans*. All stages except egg are infective (motile).

ii) **SEMI ENDOPARASITES**: These include nematodes that partially penetrate the root. E.g. *Rotylenchus*, *Helicotylenchus* (both are spiral nematode), *Criconemella* (ring nematodes) and *Paratylenchus* (pin nematodes).

iii) **ECTOPARASITES**: These are nematodes which feed on root with the aid of their long stylets but without themselves entering the root. E.g. *Xiphinema*, *Longidorus*, *Trichodorus*. They parasitize the epidermal layer of the roots.

b) **STEM, LEAF & SEED NEMATODE**: Aerial (above ground) parasites. This include rice nematodes (Aphelenchoides besseyi, A. oryzea etc which are folial nematodes. Anguina, tritici (wheat gall nematodes) are seed borne nematode & 1st to be studied. Stem nematode include Ditylenchus dipsaci.

Symptoms of Nematode Infestation

Symptoms are often observed both above and below the ground.

Above-ground symptoms include stunted growth, yellowing of leaves (chlorosis), wilting, and poor yield. These symptoms are often mistaken for nutrient deficiency.

Below-ground symptoms include root galls (swellings), lesions, rotting, and reduced root systems. In root-knot nematodes, characteristic knots or swellings are visible on roots.

Damage Caused by Nematodes

Nematodes damage crops by interfering with water and nutrient uptake. This weakens plants and reduces their productivity.

They also create entry points for secondary infections by fungi and bacteria, thereby increasing disease incidence.

Severe infestation can lead to total crop failure, especially in susceptible crops.

3.4 Methods of Nematode Control

Effective nematode control requires an integrated approach.

Cultural Control

Cultural methods include crop rotation with non-host crops, use of resistant varieties, proper sanitation, and fallowing. These practices help reduce nematode populations in the soil.

Biological Control

Biological control involves the use of natural enemies such as nematode-trapping fungi and bacteria that attack nematodes.

Chemical Control

Chemical control involves the use of nematicides to kill nematodes. However, these chemicals are often expensive and may have environmental and health risks, so their use should be carefully managed.

Physical Control

Soil solarization, which involves covering the soil with plastic to increase temperature, can help reduce nematode populations.

Integrated Nematode Management

This approach combines cultural, biological, and chemical methods to achieve effective and sustainable control.

4.0 IDENTIFICATION OF INSECT PESTS OF CROPS AND METHODS OF CONTROL

Insect pests are among the most destructive agents affecting crop production worldwide. In tropical agricultural systems such as those found in Nigeria, insect pests thrive due to favorable climatic conditions, leading to significant yield losses. Understanding insect biology, their modes of feeding, types of damage, and effective control strategies is essential for sustainable crop production.

Taxonomy of Insect

Kingdom: Animalia

Phylum: Arthropoda (Jointed legs, exoskeleton)

Subphylum: Hexapoda (6 legs)

Class: Insecta

Order: Coleoptera: Beetles

Lepidoptera: Butterflies & Moths

Diptera: True Flies

Hymenoptera: Ants, Bees, Wasps

Hemiptera: True bugs

Orthoptera: Grasshoppers & Crickets

Isoptera: Termites

Odonata: Dragon flies & Damselflies

Blattodea: Cockroaches

4.1 Characteristic Features of a Typical Insect

A typical insect belongs to the class Insecta and possesses distinct morphological features that differentiate it from other arthropods.

The insect body is divided into three main regions: the head, thorax, and abdomen. The head contains sensory organs such as compound eyes, antennae, and mouthparts used for feeding. The thorax bears three pairs of jointed legs and usually one or two pairs of wings, which enable movement. The abdomen contains vital organs responsible for digestion, reproduction, and excretion.

Insects possess an exoskeleton made of chitin, which provides protection and structural support. They also have an open circulatory system and breathe through spiracles connected to a network of tracheae.

These structural features contribute to the adaptability and survival of insects in diverse environments.

4.2 Life Cycle of Insects and Metamorphosis

Insects undergo development through a series of stages known as metamorphosis. This process can be classified into two main types: complete metamorphosis and incomplete metamorphosis.

Complete Metamorphosis (Holometabolous Development)

In this type, the insect passes through four distinct stages: egg, larva, pupa, and adult. The larval stage is usually the most destructive, as it actively feeds on plant tissues.

Examples include butterflies, moths, beetles, and flies. For instance, caterpillars (larvae of moths and butterflies) cause extensive damage to crops by feeding on leaves.

Incomplete Metamorphosis (Hemimetabolous Development)

In incomplete metamorphosis, the insect develops through three stages: egg, nymph, and adult. The nymph resembles a smaller version of the adult and gradually develops into a mature insect.

Examples include grasshoppers, locusts, aphids, and termites. Both nymphs and adults may cause damage to crops.

Understanding insect life cycles is important because control measures are often more effective at specific stages of development.

Explanation of Insect Mouthparts

The structure of insect mouthparts determines how they feed and the type of damage they cause.

Biting and chewing mouthparts are found in insects such as grasshoppers and caterpillars. These insects cut and chew plant tissues, leading to defoliation.

Piercing and sucking mouthparts are found in insects such as aphids and mosquitoes. These insects pierce plant tissues and suck sap, weakening the plant and transmitting diseases.

Some insects possess boring mouthparts that allow them to tunnel into stems, fruits, or roots, while others have cutting mouthparts that sever seedlings at the base.

4.3 Nature of Damage Caused by Insect Pests

Insect pests damage crops in different ways depending on their feeding habits.

Biting and Chewing

Insects with chewing mouthparts feed on leaves, stems, and fruits. This results in holes, defoliation, and reduced photosynthetic capacity. Severe attacks can lead to complete destruction of foliage.

Sucking and Piercing

Sap-sucking insects extract plant juices, leading to yellowing, wilting, and stunted growth. They may also secrete honeydew, which promotes the growth of sooty mold.

Boring

Boring insects penetrate plant tissues such as stems, roots, or fruits. This disrupts nutrient transport and weakens plant structure, often causing lodging or breakage.

Cutting

Cutting insects, such as cutworms, attack young seedlings by cutting them at ground level, resulting in plant death and poor crop establishment.

The type of damage often determines the appropriate control method.

4.4 Common Crop Pests and the Plants They Damage

In Nigeria, several insect pests commonly affect crops.

Grasshoppers and locusts attack cereals such as maize and millet, causing severe defoliation. Aphids infest vegetables and legumes, sucking sap and transmitting viral diseases. Stem borers affect crops like maize and rice by tunneling into stems.

Weevils attack stored grains such as maize and rice, causing post-harvest losses. Termites damage roots and stems of crops like cassava and sugarcane. Caterpillars feed on leaves of vegetables and tree crops, reducing productivity.

Each pest has specific host plants and damage patterns, which are important for identification and control.

4.5 Methods of Controlling Insect Pests

Effective insect pest control involves the use of multiple strategies to reduce pest populations to acceptable levels.

Cultural Control

Cultural methods include practices such as crop rotation, early planting, proper spacing, and destruction of crop residues. These practices reduce pest habitats and interrupt their life cycles.

Biological Control

Biological control involves the use of natural enemies such as predators, parasitoids, and pathogens to control insect pests. For example, ladybird beetles feed on aphids, while parasitic wasps attack caterpillars.

Chemical Control

Chemical control involves the use of insecticides to kill pests. This method is effective but should be used carefully to avoid resistance, environmental pollution, and harm to beneficial organisms.

Mechanical Control

Mechanical methods include hand-picking of insects, use of traps, and installation of barriers such as nets. These methods are simple but may be labor-intensive.

Quarantine Control

Quarantine measures prevent the introduction and spread of insect pests by regulating the movement of plant materials across regions.

Integrated Control (Integrated Pest Management – IPM)

Integrated Pest Management combines different control methods to achieve effective and sustainable pest control. It emphasizes monitoring pest populations and using chemicals only when necessary.

3.6 Mode of Action of Chemical Control (Contact and Systemic Insecticides)

Chemical insecticides act in different ways depending on their mode of action.

Contact insecticides kill insects when they come into direct contact with the chemical. They affect the insect's nervous system or outer body and act quickly.

Systemic insecticides are absorbed by the plant and transported through its tissues. When insects feed on the plant, they ingest the chemical and are killed. These are particularly effective against sucking insects.

Understanding the mode of action helps in selecting the appropriate insecticide for specific pests.

3.7 Procedure and Safety Precautions in Chemical Control

The use of chemical insecticides requires strict adherence to safety guidelines to protect human health and the environment.

Before application, the correct insecticide and dosage must be selected based on the target pest. Equipment such as sprayers should be properly calibrated.

Protective clothing, including gloves, masks, and boots, should be worn during application to prevent exposure. Spraying should be done under suitable weather conditions to avoid drift and contamination.

After application, equipment should be cleaned properly, and leftover chemicals should be stored safely. Empty containers should not be reused but disposed of according to safety regulations.

Farmers should also observe pre-harvest intervals to ensure that pesticide residues do not remain on crops at the time of consumption.

Conclusion

The identification and control of insect pests are essential components of crop protection. By understanding insect structure, life cycles, feeding habits, and types of damage, farmers can accurately identify pests and apply appropriate control measures. The integration of cultural, biological, mechanical, and chemical methods ensures effective pest management while minimizing environmental impact. Safe and responsible use of insecticides further enhances sustainable agricultural production.

5.0 IDENTIFICATION OF WEEDS AND METHODS OF CONTROL

Weeds are one of the major biological constraints to crop production, especially in tropical agricultural systems such as those in Nigeria. They compete aggressively with cultivated crops for essential growth resources and often result in significant yield reduction. Understanding weeds, their characteristics, effects on crops, and control methods is fundamental to effective crop production and farm management.

5.1 Understanding Weeds in Relation to Crop Production

A weed can be defined as any plant growing where it is not wanted. In agricultural systems, weeds are plants that grow among cultivated crops and interfere with their growth, development, and productivity.

Weeds are not inherently harmful by nature; however, their presence in crop fields makes them undesirable because they compete with crops for nutrients, water, sunlight, and space. In many cases, weeds are more vigorous and adaptable than cultivated crops, giving them a competitive advantage.

Weeds can be classified based on their life cycle into annual weeds (which complete their life cycle in one season), biennial weeds (which take two seasons), and perennial weeds (which live for several years). They can also be classified based on morphology into grasses, sedges, and broad-leaved weeds.

In crop production systems, weeds are considered a major limiting factor because they establish quickly, grow rapidly, and produce large quantities of seeds, making them difficult to control.

5.2 Discussion of Weeds as They Relate to Crop Production

Weeds have a direct and indirect impact on crop production. Their presence in farmland affects crop performance in several important ways.

Competition for Resources

Weeds compete with crops for essential resources such as nutrients, water, light, and space. Since many weeds grow faster and establish earlier than crops, they often outcompete crops, especially during the early stages of growth. This competition leads to reduced crop vigor and lower yields.

Reduction in Crop Yield and Quality

Weeds can significantly reduce both the quantity and quality of crop produce. For example, in cereals like maize and rice, heavy weed infestation can cause drastic yield losses. Weeds may also contaminate harvested produce, reducing its market value.

Harboring Pests and Diseases

Weeds often serve as alternate hosts for pests and disease-causing organisms. They provide shelter and breeding grounds for insects and pathogens, which can later attack cultivated crops.

Interference with Farm Operations

Weeds make farm operations such as planting, weeding, irrigation, and harvesting more difficult and labor-intensive. Dense weed growth can hinder the use of farm machinery and increase production costs.

Allelopathic Effects

Some weeds release toxic substances (allelochemicals) into the soil that inhibit the growth of nearby crops. This phenomenon, known as allelopathy, further reduces crop productivity.

5.3 Examples of Common Weeds in Nigeria

Common weeds in Nigerian agriculture include spear grass (*Imperata cylindrica*), goat weed (*Ageratum conyzoides*), Siam weed (*Chromolaena odorata*), nut grass (*Cyperus rotundus*), and pigweed (*Amaranthus spp.*). These weeds are highly competitive and difficult to control.

5.4 Identification of Weeds

Weed identification is an important step in weed management. It involves recognizing weeds based on their physical characteristics such as leaf shape, stem structure, root system, and growth habit.

Grasses have narrow leaves with parallel veins and fibrous roots. Sedges resemble grasses but have triangular stems and grow in moist conditions. Broad-leaved weeds have wider leaves with net-like veins and are often easier to distinguish.

Accurate identification helps in selecting appropriate control methods, as different weeds respond differently to management practices.

5.5 CLASSIFICATION OF WEEDS

Classification of Weeds and Basis for Classification

Weeds are classified based on several criteria, including growth habit, life cycle, and habitat. This classification helps in understanding their biology and selecting appropriate control methods.

Classification Based on Growth Habit

Weeds are grouped into grasses, sedges, and broad-leaved weeds.

Grasses are monocotyledonous plants with narrow leaves, parallel venation, and fibrous root systems. They usually have hollow stems with nodes and internodes. Examples include spear grass (*Imperata cylindrica*).

Sedges resemble grasses but can be distinguished by their solid, triangular stems and preference for moist environments. An example is nut grass (*Cyperus rotundus*).

Broad-leaved weeds are dicotyledonous plants with wide leaves and net-like venation. They include many common weeds such as pigweed (*Amaranthus spp.*) and goat weed (*Ageratum conyzoides*).

Classification Based on Life Cycle

Weeds can be classified into annuals, biennials, and perennials.

Annual weeds complete their life cycle within one growing season. They reproduce mainly by seeds and are often easier to control.

Biennial weeds complete their life cycle in two years, producing vegetative growth in the first year and flowering in the second year.

Perennial weeds live for more than two years and can reproduce through seeds as well as vegetative structures such as rhizomes, stolons, or tubers. These weeds are the most difficult to control.

Classification Based on Habitat

Weeds are also classified based on where they grow.

Terrestrial weeds grow on land and are common in crop fields. Aquatic weeds grow in water bodies such as ponds and irrigation canals. Semi-aquatic weeds grow in waterlogged or swampy conditions.

5.6 Effects of Weeds on Crop Plants

Weeds have several harmful effects on crops, which significantly reduce agricultural productivity.

Weeds compete with crops for essential resources such as nutrients, water, sunlight, and space. This competition is most severe during the early stages of crop growth.

They reduce crop yield by limiting the availability of nutrients and light required for photosynthesis. In severe cases, weeds can cause total crop failure.

Weeds also reduce the quality of harvested produce by contaminating it with seeds, debris, or toxic substances.

Another important effect is that weeds act as hosts for pests and disease-causing organisms, thereby increasing the risk of infestation in crops.

Some weeds release harmful chemicals into the soil, inhibiting crop growth through allelopathy.

Weeds also interfere with farm operations, making planting, weeding, and harvesting more difficult and costly.

5.7 Methods of Weed Control

Effective weed control involves the use of various strategies aimed at reducing weed population and minimizing their impact on crops.

Cultural Methods

Cultural weed control involves adopting good agronomic practices that suppress weed growth. These include crop rotation, cover cropping, mulching, proper spacing, and timely planting.

For example, planting fast-growing crops can help shade out weeds, while mulching prevents weed seed germination by blocking sunlight.

Mechanical Methods

Mechanical control involves the physical removal or destruction of weeds. This includes hand weeding, hoeing, slashing, and tillage.

These methods are effective, especially for small-scale farming, but can be labor-intensive and time-consuming.

Chemical Methods

Chemical control involves the use of herbicides to kill or suppress weeds. Herbicides can be selective (targeting specific weeds without harming crops) or non-selective (killing all vegetation).

They can also be classified as pre-emergence (applied before weed germination) or post-emergence (applied after weeds have emerged).

Proper application and dosage are essential to avoid crop damage and environmental pollution.

Biological Methods

Biological control involves the use of natural enemies such as insects, fungi, or grazing animals to control weeds. This method is environmentally friendly but requires careful management.

Preventive Methods

Preventive measures aim to stop weeds from entering or spreading within a field. These include using clean seeds, preventing the spread of weed seeds through machinery, and controlling weeds before they produce seeds.

Integrated Weed Management

Integrated weed management combines different control methods to achieve effective and sustainable weed control. It involves using cultural, mechanical, biological, and chemical methods in a coordinated manner.

5.8 Classification of Herbicides and Methods of Application

Herbicides can be classified based on their selectivity, timing, and mode of application.

Classification Based on Selectivity

Selective herbicides kill specific types of weeds without harming crops. Non-selective herbicides kill all plants they come into contact with.

Classification Based on Time of Application

Pre-emergence herbicides are applied before weed seeds germinate, while post-emergence herbicides are applied after weeds have emerged.

Methods of Herbicide Application

Herbicides can be applied through various methods.

Broadcast application involves spreading the herbicide uniformly over the field. Band application involves applying herbicides in narrow strips along crop rows. Spot treatment targets specific weed-infested areas.

Foliar application involves spraying herbicides directly onto weed leaves, while soil application targets weeds through the soil.

The method chosen depends on the type of weed, crop, and farming system.

5.9 Modes of Action of Herbicides

Herbicides control weeds by interfering with essential physiological processes in plants.

Some herbicides inhibit photosynthesis, preventing plants from producing food. Others disrupt cell division and growth, leading to plant death.

Certain herbicides affect amino acid synthesis, while others interfere with lipid production or enzyme activity.

Systemic herbicides are absorbed and transported within the plant, affecting internal processes, while contact herbicides destroy only the parts of the plant they touch.

Understanding herbicide modes of action helps prevent misuse and resistance development.

5.10 Factors Affecting Effectiveness of Herbicides

The effectiveness of herbicides depends on several factors.

Herbicide rate and concentration are critical; too little may be ineffective, while too much may damage crops and the environment.

Environmental factors such as rainfall, temperature, humidity, and wind affect herbicide performance. For example, rain can wash away herbicides, while wind can cause drift.

Soil properties such as texture, organic matter content, and pH also influence herbicide activity.

The growth stage of weeds is important, as younger weeds are generally more susceptible to herbicides.

Proper timing and method of application are essential for achieving maximum effectiveness.

5.11 Precautionary Measures and Hazards of Herbicide Use

The use of herbicides requires careful handling to avoid hazards to humans, crops, and the environment.

Farmers should wear protective clothing such as gloves, masks, and boots during application to prevent exposure.

Herbicides should be stored in labeled containers and kept away from children and food items.

Application equipment should be properly calibrated to ensure correct dosage and avoid wastage.

Empty containers should be disposed of safely and not reused for domestic purposes.

Herbicide misuse can lead to environmental pollution, contamination of water sources, and harm to non-target organisms, including beneficial plants and animals.

There is also a risk of herbicide resistance when the same herbicide is used repeatedly over time.

6.0 IDENTIFICATION OF VERTEBRATE PESTS OF CROPS AND METHODS OF CONTROL

Crop production is affected not only by insects, diseases, and weeds, but also by other harmful organisms such as nematodes and vertebrate pests. These organisms cause serious damage to crops, leading to reduced yield, poor quality produce, and economic losses. Understanding their biology, modes of attack, and control measures is essential for effective crop protection, particularly in tropical agricultural systems such as those in Nigeria.

6.0 VERTEBRATE PESTS OF CROPS

6.1 Common Vertebrate Pests of Crops

Vertebrate pests are animals with backbones that damage crops. Common vertebrate pests include:

- Rodents (rats, mice, squirrels)
- Birds (weaver birds, sparrows)
- Monkeys
- Grasscutters

- Antelopes and other wild animals

These pests are particularly problematic in both field and storage conditions.

6.2 Crops Commonly Affected by Vertebrate Pests

Different vertebrate pests attack different crops.

Rodents commonly damage stored grains such as maize, rice, and sorghum, as well as field crops like cassava and yam.

Birds, especially weaver birds, attack cereal crops such as rice, millet, and sorghum by feeding on grains.

Monkeys and grasscutters damage crops such as maize, banana, plantain, and cassava.

These pests often target crops at specific growth stages, particularly during fruiting and harvesting.

6.3 Nature of Damage Caused by Vertebrate Pests

Vertebrate pests cause damage in several ways.

Rodents gnaw on seeds, stems, and storage materials, leading to direct loss and contamination of produce.

Birds feed on grains and fruits, reducing yield and quality.

Monkeys and larger animals uproot plants, break stems, and consume large portions of crops.

In addition to direct feeding damage, vertebrate pests also cause indirect losses through contamination with droppings and urine, making produce unfit for consumption.

6.4 Methods of Controlling Vertebrate Pests

Effective control of vertebrate pests involves a combination of methods.

Mechanical Control (Traps and Barriers)

Traps are commonly used to capture rodents and other small animals. Physical barriers such as fences and nets are used to prevent entry into farms.

Chemical Control (Baits and Poisons)

Poisoned baits are used to control rodents. These must be used carefully to avoid harming non-target animals and humans.

Cultural Control

Proper field sanitation, timely harvesting, and proper storage reduce pest attraction. Clearing bushes around farms also reduces hiding places for pests.

Biological Control

Natural predators such as cats and birds of prey help control rodent populations.

Scaring Devices

Farmers use scarecrows, noise-making devices, and reflective materials to deter birds.

Integrated Pest Management

Combining different control methods provides the most effective and sustainable solution for managing vertebrate pests.